

Campus Services Portal - REST API & Swagger / OpenAPI Specification

Executive Overview

The **Campus Services Portal Backend** is built on **ASP.NET Core 8 Web API**. It provides a fully documented, RESTful, resource-oriented API utilizing standard HTTP verbs (`GET` , `POST` , `PUT` , `DELETE`), standard status codes, and JWT Bearer token authentication.

Interactive Swagger UI Access

The backend provides live interactive OpenAPI documentation through Swashbuckle: * **Swagger UI URL:**

`http://localhost:5000/swagger` OR `https://localhost:5001/swagger` * **Raw OpenAPI Specification:**

`http://localhost:5000/swagger/v1/swagger.json`

Authentication in Swagger UI:

1. Call `POST /api/auth/login` with your credentials.
2. Copy the returned `data.token` string.
3. Click the green **Authorize**  button at the top right of the Swagger UI.
4. Enter: `Bearer <your_token_here>` and click **Authorize**.
5. All authorized endpoints will now automatically receive the JWT in the `Authorization` header.

BRD Section 4 Functional Modules - Complete CRUD Endpoints Traceability Matrix

This table provides direct 1-to-1 evidence that **every CRUD endpoint specified in Section 4 of the BRD is implemented, secured, and returning standard HTTP status codes.**

BRD MODULE	BRD ENDPOINT & HTTP METHOD	ASP.NET CORE CONTROLLER	SECURITY & AUTHORIZATION	SUCCESS CODES	ERROR CODES	BRD BUSINESS RULE ENFORCED
Module 1: Student Profile	<code>POST /api/students/register</code>	<code>StudentsController.Register / AuthController.Register</code>	Public	<code>201 Created</code>	<code>400, 409</code>	Format validation; unique email & index; automatic student profile initialization.
Module 1: Student Profile	<code>GET /api/students/{id}</code>	<code>StudentsController.GetById</code>	<code>[Authorize]</code> (Student own profile / Admin)	<code>200 OK</code>	<code>401, 404</code>	Returns complete student profile, contact numbers, addresses, and faculty ID.
Module 1: Student Profile	<code>PUT /api/students/{id}</code>	<code>StudentsController.Update</code>	<code>[Authorize]</code> (Student own profile / Admin)	<code>200 OK</code>	<code>400, 401, 404</code>	Updates contact details; mobile OTP verification check; FormArray persistence.
Module 1: Student Profile	<code>GET /api/students?search=&faculty=</code>	<code>StudentsController.GetAll</code>	<code>[Authorize(Roles="Admin,SuperAdmin")]</code>	<code>200 OK</code>	<code>401, 403</code>	Admin-only searchable and filterable directory with pagination metadata.

BRD Module	BRD Endpoint & HTTP Method	ASP.NET Core Controller	Security & Authorization	Success Codes	Error Codes	BRD Business Rule Enforced
Module 1: Student Profile	DELETE /api/students/{id}	StudentsController.Deactivate	[Authorize(Roles="SuperAdmin")]	200 OK	400, 401, 403, 404	Rule 11: Rejects deactivation with 400 if active hostel, lab, event, certificate, or unpaid fees exist.
Module 2: Hostel Accommodation	POST /api/hostel-applications	HostelApplicationsController.Create	[Authorize(Roles="Student")]	201 Created	400, 401, 409	One active application per student; term end date must follow start date.
Module 2: Hostel Accommodation	GET /api/hostel-applications/student/{id}	HostelApplicationsController.GetByStudent	[Authorize] (Student self / Admin)	200 OK	401, 404	Returns authenticated student's current and past hostel applications.
Module 2: Hostel Accommodation	GET /api/hostel-applications?status=	HostelApplicationsController.GetAll	[Authorize(Roles="Admin,SuperAdmin")]	200 OK	401, 403	Admin review list with status filtering (Pending, Approved, Rejected).
Module 2: Hostel Accommodation	PUT /api/hostel-applications/{id}/status	HostelApplicationsController.UpdateStatus	[Authorize(Roles="Admin,SuperAdmin")]	200 OK	400, 401, 403, 404	Admin approval or rejection with reason; triggers automated notification.
Module 2: Hostel Accommodation	PUT /api/hostel-applications/{id}/assign-room	HostelApplicationsController.AssignRoom	[Authorize(Roles="Admin,SuperAdmin")]	200 OK	400, 401, 403, 404	Allocates specific room bed after approval; verifies room capacity ceiling.
Module 3: Lab Reservation	GET /api/labs	LabsController.GetAll	[Authorize]	200 OK	401	Lists Computer and Science laboratories with layout specifications.
Module 3: Lab Reservation	GET /api/labs/{id}/slots?date=	LabsController.GetSlots	[Authorize]	200 OK	400, 401, 404	Time slot list for chosen date; returns real-time slot and seat availability.
Module 3: Lab Reservation	POST /api/lab-bookings	LabBookingsController.Create	[Authorize(Roles="Student")]	201 Created	400, 401, 409	Concurrency Guard: Rejects with 409 Conflict if workstation/slot is already taken.
Module 3: Lab Reservation	DELETE /api/lab-bookings/{id}	LabBookingsController.Cancel	[Authorize] (Owner student / Admin)	200 OK	400, 401, 404	Student reservation cancellation; immediately frees slot for other students.
Module 3: Lab Reservation	GET /api/lab-bookings/student/{id}	LabBookingsController.GetByStudent	[Authorize] (Owner student / Admin)	200 OK	401, 404	Returns student booking history and upcoming active reservations.
Module 4: Event Registration	GET /api/events	EventsController.GetAll	[Authorize]	200 OK	401	Lists university events with date, venue, capacity, and registration status.
Module 4: Event Registration	POST /api/event-registrations	EventsController.Register	[Authorize(Roles="Student")]	201 Created	400, 401, 409	Rule: Blocks duplicate registration (409 Conflict); blocks registration if capacity is full (400).

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Module 4: Event Registration	DELETE /api/event-registrations/{id}	EventsController.CancelRegistration	[Authorize] (Owner student / Admin)	200 OK	400, 401, 404	Cancels event pass; decrements registered count and frees event seat.
Module 4: Event Registration	GET /api/event-registrations/student/{id}	EventsController.GetMyRegistrations	[Authorize] (Owner student / Admin)	200 OK	401, 404	Lists authenticated student's registered upcoming and past event passes.
Module 5: Complaint Management	GET /api/complaint-categories	ComplaintCategoriesController.GetAll	[Authorize]	200 OK	401	Master list of complaint categories (Maintenance, Hostel, Academic).
Module 5: Complaint Management	POST /api/complaints	ComplaintsController.Create	[Authorize(Roles="Student")]	201 Created	400, 401	Submits complaint ticket with category, urgency, and description.
Module 5: Complaint Management	GET /api/complaints/student/{id}	ComplaintsController.GetByStudent	[Authorize] (Owner student / Admin)	200 OK	401, 404	Lists student's lodged tickets and administrative status replies.
Module 5: Complaint Management	GET /api/complaints?status=	ComplaintsController.GetAll	[Authorize(Roles="Admin,SuperAdmin")]	200 OK	401, 403	Admin desk triage view filtered by status (Pending, In Progress, Resolved).
Module 5: Complaint Management	PUT /api/complaints/{id}/status	ComplaintsController.UpdateStatus	[Authorize(Roles="Admin,SuperAdmin")]	200 OK	400, 401, 403, 404	Status transition (Pending -> In Progress -> Resolved) with resolution notes.
Module 6: Certificate Requests	POST /api/certificate-requests	CertificateRequestsController.Create	[Authorize(Roles="Student")]	201 Created	400, 401, 409	Submits certificate request (Bonafide, Transcript, Completion Letter) with copies (1-5).
Module 6: Certificate Requests	GET /api/certificate-requests/student/{id}	CertificateRequestsController.GetByStudent	[Authorize] (Owner student / Admin)	200 OK	401, 404	Lists authenticated student's certificate requests and tracking notes.
Module 6: Certificate Requests	GET /api/certificate-requests?status=	CertificateRequestsController.GetAll	[Authorize(Roles="Admin,SuperAdmin")]	200 OK	401, 403	Admin queue of pending requests filterable by status.
Module 6: Certificate Requests	PUT /api/certificate-requests/{id}/status	CertificateRequestsController.UpdateStatus	[Authorize(Roles="Admin,SuperAdmin")]	200 OK	400, 401, 403, 404	Approves, issues, or rejects request; sets tracking reference and notifies student.
Module 7: Fee Payment (Simulation)	GET /api/fee-payments/student/{id}	BillingController.GetMyFees	[Authorize] (Owner student / Admin)	200 OK	401, 404	Retrieves student fee assignments, paid ledger, and outstanding balances.
Module 7: Fee Payment (Simulation)	POST /api/fee-payments/{id}/pay	BillingController.PayFee	[Authorize(Roles="Student")]	200 OK	400, 401, 404, 409	Simulates online payment; generates receipt number; blocks duplicate payments.

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Module 7: Fee Payment (Simulation)	GET /api/fee-payments/{id}/receipt	BillingController.GetReceipt	[Authorize] (Owner student / Admin)	200 OK	401, 404	Generates official payment receipt verification metadata.
Module 8: Notifications	GET /api/notifications/student/{id}	InternalNotificationsController.GetMy	[Authorize] (Owner user)	200 OK	401	Centralized system notifications for status updates across all modules.
Module 8: Notifications	PUT /api/notifications/{id}/read	InternalNotificationsController.MarkAsRead	[Authorize] (Owner user)	200 OK	401, 404	Marks individual notification as read; updates unread count.

Detailed API Endpoints Catalog by Functional Module

1. Authentication & Security Controller (/api/auth , /api/password)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS CODES	DESCRIPTION
POST	/api/auth/register	Public	StudentRegisterDto	201 Created , 400 Bad Request , 409 Conflict	Registers a new student account with index validation
POST	/api/auth/login	Public	LoginRequestDto	200 OK , 400 Bad Request , 401 Unauthorized	Authenticates user credentials and issues JWT token pair
POST	/api/auth/refresh-token	Public	RefreshTokenRequestDto	200 OK , 400 Bad Request , 401 Unauthorized	Issues a new access token using a valid refresh token
POST	/api/auth/revoke-token	Authenticated	RevokeTokenRequestDto	200 OK , 400 Bad Request	Revokes the active refresh token on sign-out
POST	/api/password/change-password	Authenticated	ChangePasswordRequestDto	200 OK , 400 Bad Request	Updates current user password with historical reuse prevention
POST	/api/password/forgot-password	Public	ForgotPasswordRequestDto	200 OK , 400 Bad Request	Generates and emails a secure password reset token
POST	/api/password/reset-password	Public	ResetPasswordRequestDto	200 OK , 400 Bad Request	Resets account password using token from email

Sample Request (POST /api/auth/login):

```
{
  "email": "student@campus.edu",
  "password": "Password123!"
}
```

Sample Response (200 OK):

```

{
  "success": true,
  "statusCode": 200,
  "message": "Authentication successful.",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refreshToken": "7c9e6679-7425-40de-944b-e07fc1f90ae7",
    "expiresAt": "2026-09-12T18:00:00Z",
    "user": {
      "id": 14,
      "email": "student@campus.edu",
      "fullName": "Dihini Jayasekara",
      "role": "Student"
    }
  }
}

```

2. Student Master & Profile Controller (/api/students , /api/studentmaster)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY / QUERY PARAMS	RESPONSE STATUS	DESCRIPTION
GET	/api/students	Admin, SuperAdmin	? search=&facultyId=&page=&pageSize=	200 OK	Retrieves paginated student directory with contact counts
GET	/api/students/{id}	Student (self), Admin	Route parameter: id	200 OK , 404 Not Found	Retrieves comprehensive profile including FacultyId
PUT	/api/students/{id}	Student (self), Admin	UpdateStudentProfileDto	200 OK , 400 Bad Request	Updates contact numbers and addresses via FormArray
DELETE	/api/students/{id}	SuperAdmin	Route parameter: id	200 OK , 400 Bad Request	Deactivates student if no active obligations exist (Rule 11)
POST	/api/studentmaster/import-csv	Admin, SuperAdmin	multipart/form-data (.csv file)	200 OK , 400 Bad Request	Bulk student intake import with validation error report

3. Hostel Management Controller (/api/hostelmanagement , /api/hostel-applications)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
GET	/api/hostelmanagement/hostels	Authenticated	None	200 OK	Lists all hostel buildings with room and bed counts
GET	/api/hostelmanagement/rooms	Authenticated	?hostelId=	200 OK	Lists rooms with current capacity and vacancy stats
GET	/api/hostel-applications	Admin, SuperAdmin	? status=&page=&pageSize=	200 OK	Retrieves hostel applications across the campus
GET	/api/hostel-applications/my	Student	None	200 OK	Retrieves the authenticated student's hostel application
POST	/api/hostel-applications	Student	CreateHostelApplicationDto	201 Created , 400 , 409	Submits a new hostel room allocation application
PUT	/api/hostel-applications/{id}/approve	Admin, SuperAdmin	HostelAllocationApprovalDto	200 OK , 400 Bad Request	Allocates a specific room bed and approves application

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
PUT	/api/hostel-applications/{id}/reject	Admin, SuperAdmin	RejectApplicationDto	200 OK , 400 Bad Request	Rejects application with reason and notifies student

4. Computer & Science Laboratories Controller (/api/labs , /api/labbookings)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY / PARAMS	RESPONSE STATUS	DESCRIPTION
GET	/api/labs	Authenticated	None	200 OK	Lists laboratories (Computer & Science)
GET	/api/labs/{id}/seats	Authenticated	Route parameter: id	200 OK	Retrieves 2D matrix workstation seats (Row, Col)
POST	/api/labs/{id}/seats	Admin, SuperAdmin	CreateLabSeatDto	201 Created , 400 , 409	Registers a physical workstation at cell address
DELETE	/api/labs/{id}/seats/{seatId}	Admin, SuperAdmin	Route params	200 OK , 400 Bad Request	Deactivates a workstation seat
GET	/api/labs/{id}/slots	Authenticated	?date=YYYY-MM-DD	200 OK	Retrieves time slots and real-time seat availability
POST	/api/labbookings	Student	CreateLabBookingDto	201 Created , 409 Conflict	Books or reserves a workstation seat with concurrency check
GET	/api/labbookings/my	Student	None	200 OK	Retrieves the authenticated student's lab bookings
DELETE	/api/labbookings/{id}	Student, Admin	Route parameter: id	200 OK , 400 Bad Request	Cancels an upcoming lab reservation

5. Events & Venues Controller (/api/events , /api/venues)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
GET	/api/events	Authenticated	?upcoming=true&search=	200 OK	Lists campus events with capacity and venue info
POST	/api/events	Admin, SuperAdmin	CreateEventDto	201 Created , 400 Bad Request	Creates a new campus event with capacity bounds
POST	/api/events/{id}/register	Student	None	200 OK , 400 , 409	Registers student for an event (blocks duplicates & full capacity)
DELETE	/api/events/{id}/register	Student	None	200 OK , 400 Bad Request	Cancels event registration ticket pass
GET	/api/events/my-registrations	Student	None	200 OK	Lists authenticated student's registered events

6. Certificate Requests Controller (/api/certificaterequests , /api/certificatetypes)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
GET	/api/certificatetypes	Authenticated	None	200 OK	Master list of certificate types (Bonafide, Transcript, etc.)
POST	/api/certificaterequests	Student	CreateCertificateRequestDto	201 Created , 400 Bad Request	Submits official document request with purpose and copy count
GET	/api/certificaterequests/my	Student	None	200 OK	Lists authenticated student's certificate requests
GET	/api/certificaterequests	Admin, SuperAdmin	? status=&page=&pageSize=	200 OK	Triage list of pending student certificate requests
PUT	/api/certificaterequests/{id}/status	Admin, SuperAdmin	UpdateCertificateStatusDto	200 OK , 400 Bad Request	Approves or issues certificate with optional tracking note

7. Complaint Management Controller (/api/complaints , /api/complaintcategories)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
GET	/api/complaintcategories	Authenticated	None	200 OK	Master list of complaint categories
POST	/api/complaints	Student	CreateComplaintDto	201 Created , 400 Bad Request	Lodges a support complaint with category and urgency
GET	/api/complaints/my	Student	None	200 OK	Lists authenticated student's logged complaints
GET	/api/complaints	Admin, SuperAdmin	? status=&priority=&page=	200 OK	Admin triage desk for complaints across faculties
PUT	/api/complaints/{id}/status	Admin, SuperAdmin	UpdateComplaintStatusDto	200 OK , 400 Bad Request	Updates ticket status (In Progress, Resolved) with feedback

8. Billing & Tuition Fee Controller (/api/billing , /api/feetypes)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
GET	/api/feetypes	Authenticated	None	200 OK	Lists active fee types (Tuition, Lab Fine, Hostel Fee)
POST	/api/feetypes	Admin, SuperAdmin	CreateFeeTypeDto	201 Created , 400 Bad Request	Creates a new system fee type definition
GET	/api/billing/ledger	Admin, SuperAdmin	? page=1&pageSize=10	200 OK	Retrieves central fee payment assignment ledger
POST	/api/billing/assign	Admin, SuperAdmin	AssignFeeRequestDto	201 Created , 200 OK , 400	Assigns fee to a single student or executes faculty bulk run
GET	/api/billing/my-fees	Student	None	200 OK	Lists fees due and paid for authenticated student

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
POST	/api/billing/payment/{id}	Student	PaymentSubmissionDto	200 OK , 400 Bad Request	Submits online payment and generates receipt

Sample Request (POST /api/billing/assign - Faculty Bulk Run):

```
{
  "studentId": null,
  "facultyId": 1,
  "feeTypeId": 2,
  "amount": 150.00,
  "billingPeriod": "2026 Semester 1",
  "description": "Semester 1 Computer Lab Maintenance Fee",
  "dueDate": "2026-10-15T00:00:00Z"
}
```

Sample Response (201 Created / 200 OK)::

```
{
  "success": true,
  "statusCode": 201,
  "message": "Bulk assignment run completed successfully. Assigned to 4 student(s).",
  "data": {
    "assignedCount": 4,
    "skippedCount": 0
  }
}
```

9. Notifications Controller (/api/internalnotifications , /api/notifications)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY / PARAMS	RESPONSE STATUS	DESCRIPTION
GET	/api/internalnotifications/my	Authenticated	None	200 OK	Retrieves real-time notifications for current user
GET	/api/internalnotifications/unread-count	Authenticated	None	200 OK	Quick endpoint for navbar badge unread counter
PUT	/api/internalnotifications/{id}/read	Authenticated	Route parameter: id	200 OK	Marks notification as read
PUT	/api/internalnotifications/read-all	Authenticated	None	200 OK	Marks all notifications read for current user

10. Audit Logs & System Administration (/api/auditlogs , /api/systemsettings , /api/adminmanagement)

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
GET	/api/auditlogs	Admin, SuperAdmin	?entityName=&action=&from=&to=	200 OK	Searchable immutable audit trail with before/after state
GET	/api/systemsettings	Authenticated	None	200 OK	Retrieves global portal configuration settings
PUT	/api/systemsettings	SuperAdmin	UpdateSystemSettingsDto	200 OK , 400 Bad Request	Updates global pagination, academic year, and flags

METHOD	ENDPOINT PATH	AUTHORIZATION	REQUEST BODY DTO	RESPONSE STATUS	DESCRIPTION
GET	/api/adminmanagement	SuperAdmin	None	200 OK	Lists administrator accounts with active/locked status
POST	/api/adminmanagement	SuperAdmin	CreateAdminRequestDto	201 Created , 400 Bad Request	Provisions a new Administrator or SuperAdmin user
PUT	/api/adminmanagement/{id}/status	SuperAdmin	ToggleAdminStatusDto	200 OK , 400 Bad Request	Activates or deactivates administrator credentials